Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Network and Data Communications

BCA — Semester 4 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Differentiate between noise and attenuation. A pure ALOHA network transmits 200 bit frames using shared channel with a 50 kbps bandwidth. What is the requirement to make this frame collision free?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is Congestion Control? How can it be handled? Explain acknowledgement policy and discarding policy.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What are the practical implications of OSI layer? Define each layer focusing on its functionality and hardware used in each layer.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Explain leaky-bucket algorithm with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Represent bit sequence 10011001 by the following wave forma. NRZ-Lb. NRZ-I

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain link state routing protocol with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain recursive resolution. What are its advantages?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Calculate the transmitted encoded frame if the message sequence is 1001001 and generator polynomial is $G(X) = x^2 + x + 1$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain the concept of TDMA with a neat diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is circuit switching? What are its advantages and disadvantages?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is the difference between DNS and DHCP? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notesa. Reliable Protocolb. Satellite Network

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
